FIITJEE

ALL INDIA TEST SERIES

FULL TEST – II

JEE (Advanced)-2020

PAPER – 1

Time Allotted: 3 Hours **Maximum Marks: 186**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-C.**

1. ***Section–A (01 – 04, 19 – 22, 37 – 40)*** *contains 12 multiple choice questions which out of 4 options have* ***only one correct answer****. Each question carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*
   Section-A (05 – 12, 23 – 30, 41 – 48) *contains 24 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for all correct answer.*
   *Full Marks : **+4** If only (all) the four option(s) is (are) chosen.*
   *Partial Marks : **+3** If all the four options are correct but ONLY three options are chosen.*
   *Partial Marks : **+2** If three or more options are correct but ONLY two options are chosen and both of which are correct options.*
   *Partial Marks : **+1** If two or more options are correct but ONLY one option is chosen and it is a correct option.*
   *Zero Marks : **0** If none of the options is chosen (i.e. the question is unanswered).*

   *Negative Marks : **–1** In all other cases.*

2. ***Section-C (13 – 18, 31 – 36, 49 – 54)*** *contains 18 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

Physics

PART – I

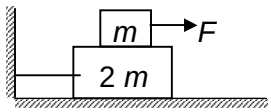
SECTION – A
(One Options Correct Type)

This section contains **04 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. The pitch of a screw gauge is 1 mm and there are 100 divisions on its circular scale. When nothing is put in between its jaws, the zero of the circular scale lies 4 divisions below the reference line. When a steel wire is placed between the jaws, two main scale divisions are clearly visible and 67 divisions on the circular scale are observed. The diameter of the wire is
(A) 2.71 mm (B) 2.67 mm
(C) 2.63 mm (D) 2.65 mm

2. Two boys are standing at the points A and B on the ground, where $AB = a$. The boy at B starts running in a direction perpendicular to AB with velocity v_1. The boy at A starts running simultaneously with constant velocity v and catches the other in time t then t is
(A) $\frac{a}{\sqrt{v^2 + v_1^2}}$ (B) $\frac{a}{v + v_1}$
(C) $\frac{a}{v - v_1}$ (D) $\frac{a}{\sqrt{v^2 - v_1^2}}$

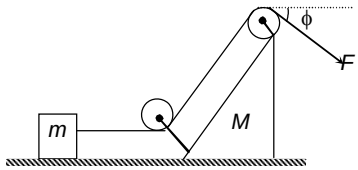
3. Two blocks of masses m & $2m$ are placed one over the other as shown in figure. The coefficient of friction between m and $2m$ is μ and between $2m$ and ground is $\mu/3$. If a horizontal force F is applied on upper block and T is tension developed in string then choose the incorrect alternative



(A) If $F = \frac{\mu}{3}mg$, $T = 0$ (B) If $F = \mu mg$, $T = 0$
(C) If $F = 2\mu mg$, $T = \mu mg/3$ (D) If $F = 3\mu mg$, $T = 0$

Space for Rough work

4. In the given arrangement pulleys and string are massless and frictionless and all surfaces are smooth. Find the magnitude of acceleration of wedge of mass M.



(A) $\frac{F\cos\varphi}{M}$

(B) $\frac{F(1-\cos\varphi)}{M}$

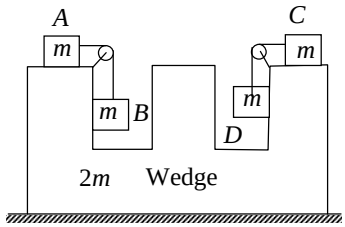
(C) $\frac{F}{m+M}$

(D) $\frac{F\cos\varphi}{m+M}$

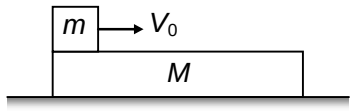
(One or More than one correct type)

This section contains **08** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

5. In the given figure, all surface are frictionless and strings and pulleys are massless, then
(A) Acceleration of block *B* is *g*/2.
(B) Acceleration of block *B* is zero.
(C) Acceleration of block wedge is zero.
(D) Acceleration of block wedge is *g*/2.



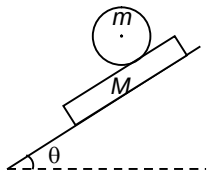
6. The coefficient of friction between the block and plank is μ and its value is such that block becomes stationary with respect to plank before it reaches the other end. Then
(A) the work done by friction on the block is negative
(B) the work done by friction on the plank is positive
(C) the net work done by friction is negative
(D) net work done by the friction is zero



Space for Rough work

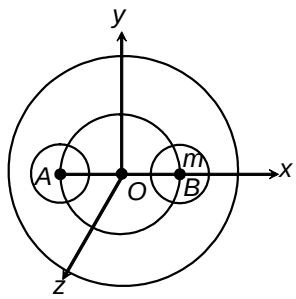
7. A chain of length L and mass per unit length ρ is piled on a horizontal surface. One end of the chain is lifted vertically with constant velocity by a force P.
(A) P as a function of height x of the end above the surface will be $\rho(gx + v^2)$.
(B) No energy will loss in this process.
(C) Work done by force will be $\frac{1}{2}\rho gL^2 + \rho v^2 L$
(D) Loss in energy $\frac{1}{2}\rho gLv^2$.

8. A solid sphere of mass m is released on the plank of mass M which lies on an inclined plane of inclination θ as shown in figure. There is sufficient frictional force between sphere and plank and the minimum value of co-efficient of friction between plank and surfaces is μ to keep the plank at rest. Then



(A) Frictional force between sphere and plank is $\frac{2}{7}mg\sin\theta$, when plank is at rest.
(B) The value of μ is $\frac{7M+2m}{7(M+m)}\tan\theta$.
(C) If there is no friction between the plank and inclined plane, then acceleration of plank is less than $g \sin \theta$.
(D) If there is no friction between plank and inclined plane, then friction force on the sphere is zero.

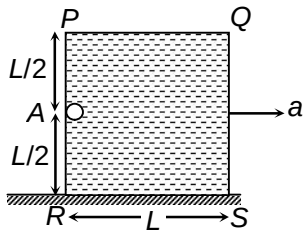
9. A solid sphere of uniform density and radius 4 unit is located with its center at the origin O of co-ordinates. Two spheres of equal radii 1 unit, with their centres at A(-2,0,0) and B (2, 0 ,0) respectively, are taken out of the solid leaving behind spherical cavities as shown in figure. Then



(A) The gravitational field due to this object at the origin is zero.
(B) The gravitational field at the point B(2, 0, 0)is zero
(C) The gravitational potential is same at all points on the circle $y^2 + z^2 = 36$
(D) The gravitational potential is same at all points on the circle $y^2 + z^2 = 4$

Space for Rough work

10. A small solid ball of density ρ is held inside at point *A* in a closed cubical container of side *L*, filled with an ideal liquid of density 4ρ as shown in the figure. Now, if the container starts moving with constant acceleration *a* horizontally and the ball is released from point *A* simultaneously, then



(A) For ball to hit the top of container at end *Q*, $a = 3g$

(B) For ball to hit the top of container at end *Q*, $a = 2g$

(C) Ball hits the top of container at end *Q* after a time $t = \sqrt{L/3g}$

(D) Ball hits the top of container at end *Q* after a time $t = \sqrt{2L/3g}$

11. A particle of mass *m* moves in a straight line. If *v* is the velocity at a distance *x* from a fixed point on the line and $v^2 = a - bx^2$, where *a* and *b* are constant, then

(A) The motion continues along the positive *x*-direction only

(B) The motion is simple harmonic

(C) The particle oscillates with a frequency equal to $\frac{\sqrt{b}}{2\pi}$

(D) The total energy of the particle is *ma*

12. A sound wave of frequency ν travels horizontally to the right. It is reflected from a large vertical plane surface moving to the left with a speed *V*. The speed of sound in the medium is *c*.

(A) The number of wave pulse striking the surface per second is $\frac{\nu(c+V)}{c}$

(B) The wavelength of the reflected wave is $\frac{c(c-V)}{\nu(c+V)}$

(C) The frequency of the reflected wave is $\frac{\nu(c+V)}{(c-V)}$

(D) The number of beats heard by a stationary listener to the left of the reflecting surface is $\frac{v\,V}{c-V}$

Space for Rough work

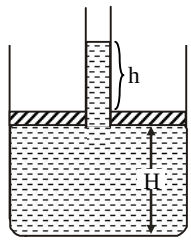
SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

13. A barometer has become faulty because some air has leaked into the space above the mercury column. The height of the mercury column is 70 cm and the length of the space above the mercury is 10 cm. The barometer tube is then pushed further down into the mercury through. The length of space above mercury reduces to 6cm and the height of the mercury column is found to decrease to 67cm. This information shows that the true atmospheric pressure must be:

14. The distance between the parallel plates of a charged condenser is d = 5cm and the intensity of the field E = 300 V cm^{-1}. A slab of dielectric constant K = 5 and 1 cm wide is inserted parallel to the plates. Determine the potential difference between the plates, after the slab is inserted. (in volt)

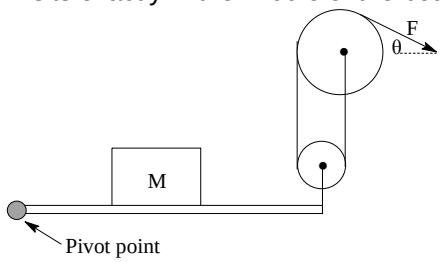
15. A piston weighing W = 3 kg has the form of a circular disc of radius R = 4 cm. The disc has a hole into which a thin - walled pipe of radius r = 1 cm in inserted. Initially the piston is at the bottom of the cylinder. What height will the piston rise to if m = 700 g of water is poured into the pipe?



16. A uniform rod of length l = 1m is free to move and rotate in gravity-free space. When an impulse is given to one end of the rod, perpendicular to its length, its centre of mass moves with velocity v = 1 m/s. What will be its angular velocity (in rad/s) about its centre of mass?

Space for Rough work

17. A pulley system is attached to a massless board as shown below. The board pivots only at the pivot point. A 10 kg mass M sits exactly in the middle of the board.



If the angle θ is 60°. The force F (in N) necessary to hold the board in the position shown is $5\times x$, find x?

18. The peak emission from a black body at a certain temperature occurs at a wavelength of 6000 Å. On increasing its temperature, the total radiation emitted is increased 16 times. These radiations are allowed to fall on a metal surface. Photoelectrons emitted by the peak radiation at higher temperature can be bought to rest by applying a potential equivalent to the excitation potential corresponding to the transition for the level n = 4 to n = 2 in the Bohr's hydrogen atom. The work function of the metal is given by $\frac{\alpha}{100}\text{eV}$ where α is the numerical constant. The value of α is found $53\times x$, find x?

[Take: he = 12420 eV-Å]

Space for Rough work

Chemistry

PART – II

SECTION – A
(One Options Correct Type)

This section contains **04 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

19. Which of the following ions is expected to be colourless?
(A) Ni^{2+} (B) Mn^{+2}
(C) Zn^{2+} (D) Cu^{2+}

20. A definite mass of a monoatomic ideal gas at 1 bar and 27°C expands against vacuum from 1.2 dm^3 to 2.4 dm^3. The change in free energy of the gas, ΔG, is (R = 0.80 bar – L/K-mol, ln 2 = 0.7)
(A) 0 (B) $-64\,\text{bar}-1$
(C) $+84\,J$ (D) $-84\,J$

21. In a galvanic cell, the salt bridge:
(A) Does not participate chemically in the cell reaction.
(B) Stops the diffusion of ions from one electrode to another.
(C) Is necessary for the occurrence of the cell reaction.
(D) Ensures mixing of the two electrolytic solutions.

22. In the system, $\text{LaCl}_3(\text{s}) + \text{H}_2\text{O}(\text{g}) + \text{heat} \rightleftharpoons \text{LaClO}(\text{s}) + 2\text{HCl}(\text{g})$, equilibrium is established. More water vapour is added to disturb the equilibrium. If the pressure of water vapour at new equilibrium is double of that at initial equilibrium, the factory by which pressure of HCl is changed is:
(A) 2 times (B) $\sqrt{2}$ times
(C) $\frac{1}{\sqrt{2}}$ times (D) 4 times

Space for Rough work

(One or More than one correct type)

This section contains **08** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

23. Which of the following information(s) is/are incorrect regarding the voids formed in three dimensional HCP of identical spheres?
(A) A tetrahedral void is formed when a sphere of the second layer is present above the triangular void of the first layer.
(B) All the triangular voids are not covered by the spheres of the second layer.
(C) Tetrahedral voids are formed when the triangular voids in the second layer lie above the triangular voids in the first layer and the triangular voids in the first layer and the triangular shapes of these voids do not overlap.
(D) Octahedral voids are formed when the triangular voids in the second layer exactly overlap with similar voids in the first layer.

24. Consider 0.1m-aqueous solutions of two solutes X and Y. The solute X behaves as a univalent electrolyte, while the solute Y dimerizes in solution. Which of the following statement(s) is/are correct regarding these solutions?
(A) The boiling point of the solution of X will be higher than that of Y.
(B) The osmotic pressure of the solution of X will be higher than that of Y.
(C) The freezing point of the solution of X will be higher than that of Y.
(D) The relative lowering of vapour pressure of both the solution will be the same.

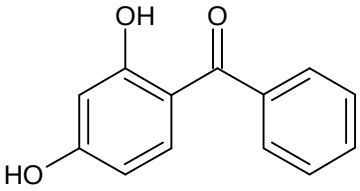
25. Select the correct statement:
(A) Chelation effect is maximum for five and six membered rings
(B) Greater the charge on the central metal cation, greater the value of $\Delta\left(\mathrm{CFSE}\right)$
(C) In complex ion $\left[\mathrm{CoF_6}\right]^{3-}$, $\mathrm{F^-}$ is a weak field ligand, so that $\Delta_{oct} < P$ (Pairing energy) and it is low spin complex.
(D) $\left[\mathrm{CoCl_2\left(NH_3\right)_2\left(en\right)}\right]^{+}$ complex ion will have four different isomers

26. Roasting of copper pyrites is done:
(A) to remove moisture
(B) to oxidise free sulphur
(C) to decompose pyrite into $\mathrm{Cu_2S}$ and FeS
(D) to remove volatile organic impurities

27. Select the incorrect statement(s):
(A) $\mathrm{Cs^+}$ is more hydrated than the other alkali metal ions
(B) Among the alkali metals Li, Na, K and Rb, lithium has the highest melting point
(C) Ionic mobility of $\mathrm{Li^+}$ is maximum among alkali metal cations
(D) Ionisation potential of Li is lower than that of Na.

Space for Rough work

28. A sunburn prevention agent must be non-toxic, non-staining, non-volatile, reasonably water soluble. It should filter of the burning rays of sunlight. Given structure is one such compound:



which method/s for synthesis of above compound starting from m-dimethoxybenzene:

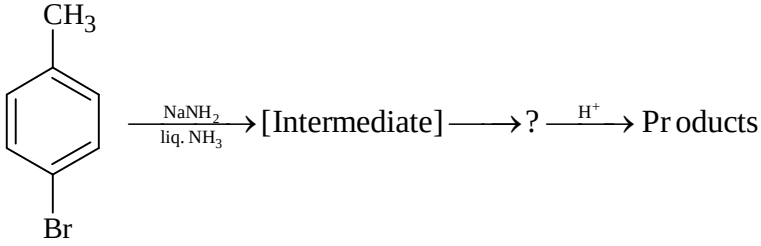
(A) (i) Ph-COCl, (ii) dil. NaOH

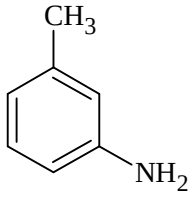
(B) (i) Conc. HI (ii) Ph-COCl, $AlCl_3$

(C) (i) Ph-COCl, $AlCl_3$ (ii) conc. HI

(D) All of these

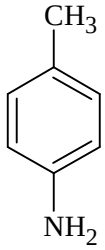
29. Choose the correct statement(s)



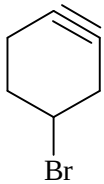
(A) 

is formed.

(B) Intermediate formed is aromatic

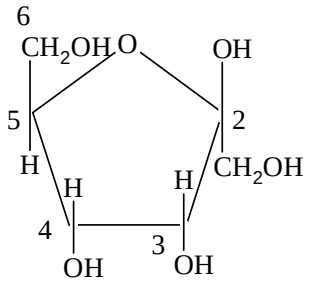
(C) 

is formed involving non-aromatic intermediate

(D) The intermedaite is 

Space for Rough work

30. Select correct statement/s for structure



(A) β–D–fructofuranose
(B) α–D–fructofuranose
(C) is a reducing sugar
(D) is a non-reducing sugar

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

31. An optically active substance, A, decomposes into optically active substances 'B' and 'C' as:

$A \xrightarrow{K=0.001\ \text{min}^{-1}} 2B + C$

The specific rotations of A, B and C are +40°, +10° and $-30°$ per mole, respectively. If initially A and C were present in $4:3$ mole ratio, the time (in min), after which the sample becomes optically inactive, is (ln 2=0.7, ln 5=1.6, ln 7=2.0, ln 13=2.5)

32. A hydrogenation reaction is carried out at 500K. If the same reaction is carried out in the presence of a catalyst at the same rate, the temperature required is 400 K. The activation energy of reaction (in kJ/mol), if the catalyst lowers the activation energy by 20 kJ/mol, is:

Space for Rough work

33. A volume of 100 ml of a liquid contained in an isolated container at a pressure of 1 bar. The pressure is steeply increases to 100 bar by which the volume of liquid is decreased by 1 ml. The change in enthalpy, ΔH, of the liquid (in bar mL)

34. At 20°C, the vapour pressure of 0.1 molal aqueous solution of urea is 0.03 mm less than that of water and the vapour pressure of 0.1 molal solution if KCl is 0.0594 mm less than that of water. The apparent percentage dissociation of KCl in water at the given temperature is (Neglect the moles of solute particles in comparison to the moles of water in both solutions).

35. If x and y are total number of electrons which are present in non-axial and axial set of d-orbitals respectively in Ni cation of $\left[\mathrm{Ni}\left(\mathrm{DMG}\right)_2\right]$, then calculate value of $\frac{2x^2}{y}$.

36. Total energy of electron in the 1st orbit of hydrogen atom is 13.6 eV/atom. Kinetic energy of electron in the 1st excited state of Li^{2+} ion in eV is:

Space for Rough work

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **04 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

37. If $f(x) = -1 + ax + a$ neither touches nor cuts the curve $f(x) = \ln x$, then minimum value of a may lies in

(A) $\left(\frac{1}{e}, \frac{1}{\sqrt{e}}\right)$ (B) $\left(e, e^2\right)$

(C) $\left(\frac{1}{\sqrt{e}}, e\right)$ (D) None of these

38. If $A(z_1), B(z_2), C(z_3)$ are vertices of a triangle such that $z_3 = \left(\frac{z_2 - iz_1}{1-i}\right)$ and $|z_1| = 3, |z_2| = 4$ and $|z_2 + i\,z_1| = |z_1| + |z_2|$, then area of triangle ABC is

(A) $\frac{5}{2}$ (B) 0

(C) $\frac{25}{4}$ (D) $\frac{25}{2}$

39. Let $A = \{1,2,3,4,5\}$ *and* $B = \{3,4,5,6,7,8,9\}$ are two sets, then the number of one-one functions $f : A \to B$ such that $|f(i) - i| \le 4 \forall i \in A$, is equal to

(A) 243 (B) 120

(C) 149 (D) 219

Space for Rough work

40. A curve is represented parametrically by the equation $x = f(t) = a^{\ln(b^t)}$ *and* $y = g(t) = b^{-\ln(a^t)}$, $a, b > 0$ and $a \neq 1, b \neq 1$ where $t \in R$ the value of $\frac{d^2y}{dx^2}$ at the point where $f(t) = g(t)$ is

(A) 0 (B) $\frac{1}{2}$

(C) 1 (D) 2

(One or More than one correct type)

This section contains **08** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

41. Let A *and* B are commutative square matrices of order three such that A is symmetric and B is skew symmetric, if $C = (A+B)^T (A+B)(A-B)^{-1}$ (where X^T *and* $|x|$ denotes transpose and determinant value of x respectively), then

(A) $A+B-C=0$ (B) $A-B-C=0$

(C) $|B|=0$ (D) $2A-C=C^T$

42. If $\lim_{n\to\infty}\left(\frac{1}{1^2}+\frac{1}{2^2}+.....+\frac{1}{n^2}\right)=\frac{\pi^2}{6}$ *and* $S=\lim_{n\to\infty}\left(\frac{1}{1^3\cdot 2^3}+\frac{1}{2^3\cdot 3^3}+\frac{1}{3^3\cdot 4^3}+....\frac{1}{n^3\cdot(n+1)^3}\right)$, then $S+\pi^2$ is equal to

(A) A non-integral number (B) An rational number

(C) A transcendental number (D) An integer

43. $f(x)$ is a differentiable function such that $f(x) = x^2 + \int_0^x e^{-t} f(x-t)dt$, then

(A) $f(x)=\frac{x^3}{3}+x$ (B) $f(x)=\frac{x^3}{3}+x^2$

(C) $f'(x)=x^2+2x$ (D) $\int_{-2}^{2}\left(f(x)-x^2\right)dx=0$

Space for Rough work

44. Which of the following is/are true?

(A) If $y = f(x)$ is a strictly monotonic function in (a,b) with $f'(x) \neq 0$, then the area bounded by $y = f(x)$, the lines $x = a, x = b$ and $y = f(c)$ where $c \in (a,b)$ is minimum when $c = \frac{a+b}{2}$.

(B) $y = f(x)$ is a strictly monotonic function in (a,b) with $f'(x) \neq 0$, then the area bounded by the lines $x = a, x = b$ and $y = f(c)$ where $c \in (a,b)$ is minimum when $c = \frac{2a+b}{3}$ or $c = \frac{a+2b}{3}$.

(C) If the area bounded by $f(x) = \frac{x^2}{3} - x^2$ and the lines $x = 0, x = 2$ and the x-axis is minimum when $a = \frac{2}{3}$

(D) If the area bounded by $f(x) = \frac{x^3}{3} - x^2$ and the lines $x = 0, x = 2$ and the x-axis is minimum when $a = \frac{28}{81}$

45. Mr. A randomly picks 3 distinct numbers from the set $\{1,2,3,.....,9\}$ and arranges them in descending order to form a three digit number. Mr. B randomly picks 3 distinct numbers from the set $\{1,2,3,....,8\}$ and also arranges them in descending order to form 3 digit numbers.

(A) Probability that Mr. A's 3 digit number is always greater then Mr. B's 3 digit number is $\frac{1}{3}$

(B) Probability that Mr. A and Mr. B has the same 3 digit numbers is $\frac{1}{84}$

(C) Probability that Mr. A's 3 digit number is greater then Mr.B's 3 digit number is $\frac{37}{56}$

(D) Probability that Mr. A's 3 digit number is greater then Mr.B's 3 digit number is $\frac{1}{3}$

Space for Rough work

46. Let $H:\frac{(x-1)^2}{9}-\frac{(y+2)^2}{4}=1$ be the hyperbola with centre at C. Now if from any point 'Q' on the asymptotes a straight line be drawn perpendicular to the transverse axis to meet the hyperbola at P and P'. Again a tangent is drawn to the hyperbola at P which meet the asymptotes at L and G, then

(A) $PQ.QP'=4$ (B) $PQ.QP'=9$

(C) area of $\Delta CLG=6$ (D) area of $\Delta CLG=8$

47. If $I_1=\int_0^{\frac{\pi}{2}}\frac{\sin(\sin x)}{\sin x}dx; I_2=\int_0^{\frac{\pi}{2}}\frac{\sin x}{x}dx; I_3=\int_0^{\frac{\pi}{2}}\frac{\sin(\tan x)}{\tan x}dx$, then which of the following is true

(A) $I_1>I_3$ (B) $I_2>I_3$

(C) $I_1>I_2$ (D) $I_1<I_2$

48. An ellipse whose major axes is parallel to x-axis is such that the segment of a focal chord are of length 1 and 3 unit. The family of lines $ax+by+c=0$ are the chords of the ellipse such that a,b,c are in $A.P.$ and bisected by the point at which they are concurrent and the equation of auxiliary circle is.

$x^2+y^2+2\alpha x+2\beta y-2\alpha+1=0$, then make all the correct alternative(s)

(A) The equation of the auxiliary circle is $x^2+y^2-2x+4y+1=0$

(B) Eccentricity of ellipse if $\frac{1}{2}$

(C) Length of major axis are $4,\sqrt{3}$

(D) Eccentricity of the ellipse is $\frac{\sqrt{3}}{2}$

Space for Rough work

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

49. The line $\frac{x-2}{3}=\frac{y+1}{2}=\frac{z-1}{-1}$ intersects the curve $xy=c^2$, in xy plane if c^2 is equal to_____

50. If $a+b+c=4$, then maximum value of $\sqrt{3a+5}+\sqrt{3b+5}+\sqrt{3c+5}$ is equal to_____ $\left(where\ a,b,c\in R^+\right)$

51. If K^{th} term is the greatest term in the square $a_n=\frac{n}{70+n^2}, n\in N$, then k is____

52. A triangle is so placed that the midpoints of its sides one placed on the co-ordinate axes, if a, b and c are the sides of the triangle, then the equation of the triangle is $\frac{x}{x_1}+\frac{y}{y_1}+\frac{z}{z_1}=1$ where $a^2+b^2+c^2=k\left(x_1^2+y_1^2+z_1^2\right)$, then the value of k is_____

53. The value of $\tan 1^\circ\tan 2^\circ+\tan 2^\circ\tan 3^\circ+.....+\tan 88^\circ\tan 89^\circ$ is equal to $\cot^2 1^\circ-n$, where n is a two digit number ab, then the value of $b-a$ is_____

54. All the three vertices of an equilateral triangle lie on the parabola $y=x^2$, and one of its sides has a slope of 2. The x-coordinates of the three vertices have a sum equal to $\frac{p}{q}$ where p and q are relatively prime positive integers. Find the value of (q-p)

Space for Rough work